Chapter 39 – More on the Electric Potential
Calculus Physics II

Conservation of the Electric Force - The electric force is a conservative force (same as gravitational force), therefore in moving a charged object in an electric field, there is no loss or gain in the total energy of the charged object over time, the energy just changes form.

- ❖ Total mechanical energy does not change over time therefore is constant.
- ❖ There is NO net work done on the object (W = 0J)
- ❖ The energy created by a charged object in an electric field is conserved.

$W = \Delta KE + \Delta PE = 0$ for Energy to be conserved!

$$\Delta KE = -\Delta PE$$

- The change in kinetic energy can be translational motion (ΔKE) on rotational motion (ΔKE_R) or both.
- The change in potential energy can be gravitational potential energy (ΔPE_g), Elastic potential Energy (ΔPE_e) or Electric Potential Energy (ΔPE_E).
- The electric potential energy can be converted to Kinetic Energy of the charged particle in the same way gravitational PE is converted to KE as the object falls.

Example: For a charged particle starting from rest (v_i = 0 m/s) in a constant Electric Field the conservation of energy and assuming no change of energy due to the gravitational force or any elastic forces gives

$$\Delta KE + \Delta PE_E = 0$$

Determining the change in translational kinetic energy and the electric potential energy

$\Delta KE = KE_f - KE_i = \frac{1}{2}mv_f^2 - 0 = \frac{1}{2}mv_f^2$

$\Delta PE_E = -q\Delta V$

substituting for the variables in the above equation $\frac{1}{2}mv_f^2 = -(-q\Delta V)$ and solving for the initial velocity

$$v_f = \sqrt{\frac{2q\Delta V}{m}}$$

- The velocity ($\vec{v}$) of the charged particle is dependent on the square root of the amount of charge and change in electric potential and inversely dependent on the mass.

Example: A charged sphere (m_{sphere} = 5 g and q = 100 μC negative) is moving with a constant velocity of 10 m/s when it enters a electric field, the sphere moves along the electric field line in the direction of the electric field, what change in potential is required to bring the sphere to a stop? The sphere enters the field at position A and comes to a stop at position B.

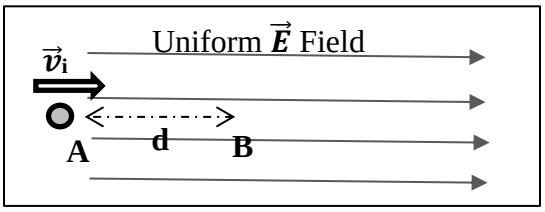
$$\Delta KE + \Delta PE_E = 0$$

Determining the change in kinetic and potential energy of the object

$\Delta KE = KE_f - KE_i = 0 - \frac{1}{2}mv_i^2$

$\Delta PE_E = q\Delta V \qquad \Delta KE = -\Delta PE$

$-\frac{1}{2}mv_i^2 = -q\Delta V$ solving for ΔV



$$\Delta V = \frac{mv_i^2}{2q} = 2500\ V$$

The electron Volt (eV) – The electron Volt is a unit of energy and is the amount the change in the potential energy of an electron when the electron moves through a potential difference of one volt.

$PE_{Elec} = q\Delta V = (1.6 \times 10^{-19}C)(1.00\ V) = 1.6 \times 10^{-19}\ J$

1 eV = 1.6 x 10^{-19} J

Absolute Potential (V) for a single point Charge- The absolute potential at a point in space a distance r is the work done against the electric force in carrying a unit positive test charge from infinity to a point r distance from the point charge q.

$$\vec{E} = \left(k\frac{q}{r^2}\right)\hat{r}$$

$$V_{absolute} = -\int_{\infty}^{r}\vec{E}\bullet d\vec{r} = -\int_{\infty}^{r}k\frac{q}{r^2}\hat{r}\bullet d\vec{r} = -kq\left[-\frac{1}{r}-\frac{1}{\infty}\right] = \frac{kq}{r}$$

$$V = \frac{kq}{r} \qquad \text{where } k = 9.0 \times 10^9\ Nm^2/C^2$$

q - is the net charge and is the charge creating the electric potential.
r - is the distance from the charge to the point in space where the potential is to be determined.

- The absolute Potential gives the change in potential or potential difference between a point in space and infinity.

The algebraic "sign" of Potential (V)

- A positive point charge creates a positive electric potential. The potential starts out large and as the radius (distance from charge) increases the potential decreases.
- A negative point charge creates a negative electric potential. The potential starts out a large and it becomes less negative as the radius is increased.

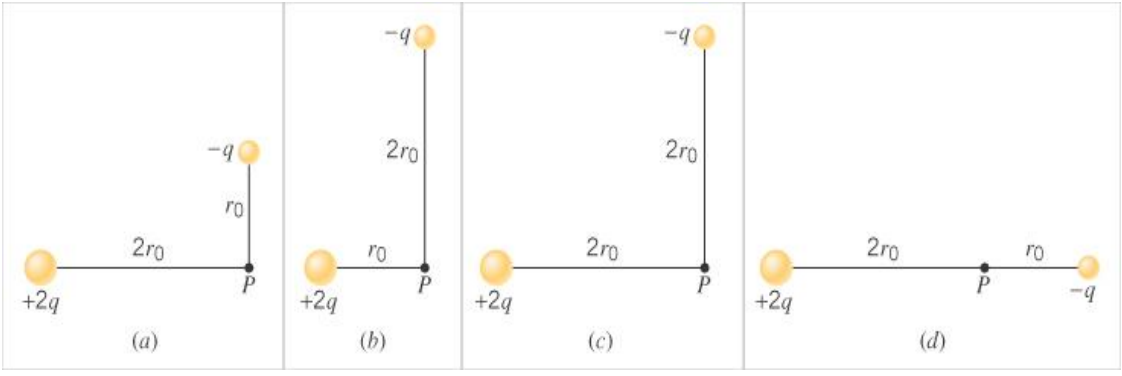


Determining Potential at a point due to many charges - Using the Superposition Principle for scalars, the total electric potential at a point in space due to two or more charges is equal to the algebraic sum of the potentials due to each charge separately at a point in space, since the electric potential is a scalar the sum is just adding numbers taking care to use the correct sign of the potential.

- Positive point charge creates an algebraically positive potential (+ V).
- Negative point charge creates an algebraically negative potential (-V).

$$V_{Total} = V_1 + V_1 + V_1 + \cdots = \frac{kq_1}{r_1} + \frac{kq_2}{r_2} + \frac{kq_3}{r_3} + \cdots$$

Example: Determine the Potential at point P for the 4 different charge distributions if q = 1 μC and r_o = 1 m?



The potential at P is due to a positive potential created by the positive 2q charge and a negative potential created by the negative q charge, potential is a scalar therefore the calculated potentials are added taking into account the "sign". Calculating each potential at point P and assigning a positive potential for the positive charge and a negative potential from the negative charge.

(a) $V_{Total} = (k2q)/2r_o + (-kq/r_o) = (+ 9000\ V) + (- 9000\ V) = 0\ V$

(b) $V_{Total} = (k2q)/r_o + (-kq/2r_o) = (+1800\ V) + (- 4500\ V) = + 13{,}500\ V$

(c) $V_{Total} = (k2q)/2r_o + (-kq/2r_o) = (+9000\ V) + (- 4500\ V) = + 4{,}500\ V$

(d) $V_{Total} = k2q/2r_o + (-kq/r_o) = (+ 9000\ V) + (- 9000\ V) = 0\ V$

Electric Potential due to a charge distribution - The electric potential at a point in space due to a continuous distribution of charge is found by integrating the contributions from each infinitesimal element of charge, denoted as dq. The electric potential is not a vector quantity therefore only the amount of potential created by each incremental dq is added to obtain the total electric potential at the point a distance r away. The general form of the electric potential due to a charge distribution is:

$$Electric\ Potential \quad V(r) = k\int \frac{1}{r}dq$$

The Electric Potential for charge distributions.

$$Electric\ Potential\ for\ a\ Line\ Charge\ V(r) = k\int \frac{\lambda}{r}dl$$

$$Electric\ Potential\ for\ a\ Surface\ Charge\ V(r) = k\int \frac{\sigma}{r}dA$$

$$Electric\ Potential\ for\ a\ Volume\ Charge\ V(r) = k\int \frac{\rho}{r}dV$$

To determine the electric potential due to a charge distribution:

1. Identify the charge distribution:
   - Determine the type of charge distribution (e.g., point charges, line of charge, surface charge, volume charge) and its geometry.

2. Divide into infinitesimal charges:
   - Break down the charge distribution into infinitesimal charge elements, dq.

4. Set up the integral:
   - Since the potential is a scalar, you can sum (integrate) the potentials from all the dq elements. The integral will be over the volume, surface, or line of the charge distribution, depending on the type of distribution.

5. Evaluate the integral:

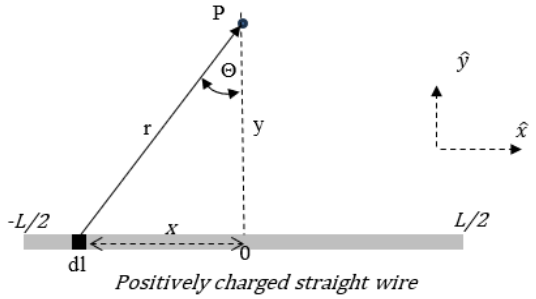
- Evaluate the integral, taking into account the geometry of the charge distribution and the relationship between dq and the integration variable(s). You may need to use coordinate systems and appropriate substitutions.

6. Determine the potential at the point:

- The result of the integration is the total electric potential at the point you are interested in.

  - Electric potential is a scalar quantity, meaning you only need to consider its magnitude and not its direction when summing the contributions from different charge elements.
  - If the charge distribution has symmetry, you can often simplify the calculation by taking advantage of that symmetry.

Example: A positively charged straight wire with a uniform linear charge density of λ, determine the electric field strength at point P, y meters above the wire and situated at the above the center of the wire of length L. Assume y and L are constant and do not change in this problem.



For a line of charge, you would express dq in terms of the linear charge density (λ) and a length element (dL = dx), so dq = λdx. The distance r would be expressed in terms of the coordinates (x, y, z) of the point and the position of the charge element. You would then integrate along the length of the line.

Determine the incremental charge on the wire.

$$dq = \lambda dl = \lambda dx$$

$$Electric\ Potential\ for\ a\ Line\ Charge\ V(r) = k\int \frac{\lambda}{r} dl$$

Deterine r in terms of the geometry of the system

$$r^2 = x^2 + y^2 \quad and\ r = \ (x^2 + y^2)^{1/2}$$

Substituting into integral

$$V(r) = k\int \frac{\lambda}{r}dl = k\lambda \int_{-L/2}^{+L/2} \frac{1}{(x^2+y^2)^{1/2}}dx$$

Evaluating the integral

$$V(r) = k\lambda \left[\ln\left(x + \sqrt{x^2+y^2}\right]_{-L/2}^{+L/2}\right.$$

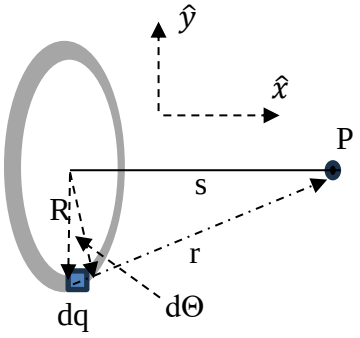
$$V(r) = k\lambda \left(\left[\ln\left(L/2 + \sqrt{(L/2)^2+y^2}\right)\right] - \left[\ln\left(-L/2 + \sqrt{(-L/2)^2+y^2}\right)\right]\right)$$

Example: A circular wire with a radius R and a uniform linear charge density of λ is shown below, determine the electric potential V at point P which is at a distance of s from the center of the disk. Determine the incremental charge on the wire.

$$dq = \lambda dL = \lambda(Rd\Theta) \text{ where } dL = Rd\Theta$$

The radius $r = \sqrt{R^2+s^2}$ *and is constant*

Electric Potential for a Line Charge $V(r) = k\int \frac{\lambda}{r}dL$



$$V(r) = k\int \frac{\lambda}{r}dL = k\lambda\int_0^{2\pi}\frac{dq}{r} = k\lambda\int_0^{2\pi}\frac{Rd\Theta}{\sqrt{R^2+s^2}} = \frac{k\lambda R}{\sqrt{R^2+s^2}}\int_0^{2\pi}d\Theta$$

$$V(r) = \frac{k\lambda R 2\pi}{\sqrt{R^2+s^2}}$$

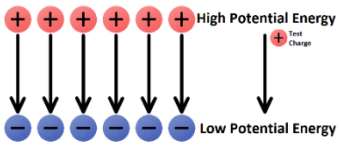
Note that when s = 0 m the electric potential at the center of the circular wire is equal to

$$V(r) = k\lambda 2\pi$$

Sources of Electric Potential – any device that create an electric field creates an electrical Potential. Some of the more common sources are listed below.

Electric generator – uses a changing magnetic field and mechanical energy to create an electric field.

Photoelectric effect – light (photons) interacts with semiconductors to create charge separation that creates an electric field.



Chemical (batteries) – uses chemical reactions to separate the charge that creates an electric field.

Friction – rubbing a material and mechanical transfer of charge to produce the separation.

Piezoelectric effect – uses pressure and a unique atomic structure of a crystalline substance to produce the charge separation that creates an electric field.

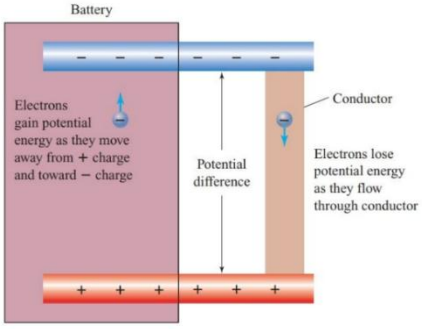
Source	Voltage (approx.)
Thundercloud to ground	10^8 V
High-voltage power line	10^5–10^6 V
Power supply for TV tube	10^4 V
Automobile ignition	10^4 V
Household outlet	10^2 V
Automobile battery	12 V
Flashlight battery	1.5 V
Resting potential across nerve membrane	10^{-1} V
Potential changes on skin (EKG and EEG)	10^{-4} V

- Creation of the electric field is the source of all our electric power sources!
- The electric field is what creates the potential energy and defines the electric potential (Voltage).
- The electric field is the source of the energy and moves the charge through the system.

The Battery and the Electric Potential - The battery does work to separate the positive and negative charge through its chemical energy. The battery must do work to moving positive charge together at one end of the battery and negative charge at the other. Because like charges repel, a force must be applied over a distance to move the same charges together, this is work which creates an electric field which in turn establishes the electric potential energy.



- The larger the battery voltage the more energy it has to perform work 1 V = 1 J/C.
- The charge separation creates an electric field; the electric field creates the force that moves the charge that does the work.



By moving the electron from a lower potential energy (high potential because potential is defined by a positive charge) to a higher potential energy (low potential) the electron now has energy that can be used to do work (run your phone or a flashlight!). It is the electric field that was created by separating the charge that gives the battery its potential energy which gives the battery its potential (voltage).

Example: Determine the Work required to move an electron (e^-) against the electric field moving from + pole to – pole of a 9 Volt battery? Using $\Delta V = - W/q$

$$W = -\Delta PE_E = q\Delta V = 1.6 \times 10^{-19}C\ (9V) = 1.44 \times 10^{-18}\ J$$

<u>Zero Potential and Ground</u> - Zero electric potential means that the electric potential energy of a charge at that point is zero relative to a chosen reference point.

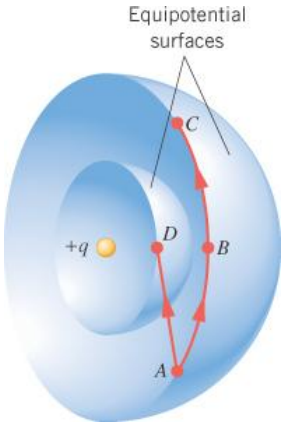
- The zero electric potential point is arbitrary the same way the zero gravitational potential energy is arbitrary. You can set any point to be zero potential.
- The Change in Potential is the only value that has real meaning.
- The zero Potential is usually set to make the problem easier or more logical.
- Zero potential is sometimes set at infinity because that is the only place where the electric field is zero in theory.
- All other potentials are measured relative to the zero Potential.
- Ground (or Earth) is defined as having zero potential. In real life, the Earth is so huge that it can absorb or supply charge without changing its own potential much, so we treat it as a stable zero.
- The Potential can be negative depending on where the ground (zero potential) is defined.
- In circuits and physics problems, ground is a special reference point that we define as having zero electric potential.
- Zero Potential does not mean that the electric field is zero!

<u>Equipotential Surface</u> – a surface on which the electric potential is the same everywhere.

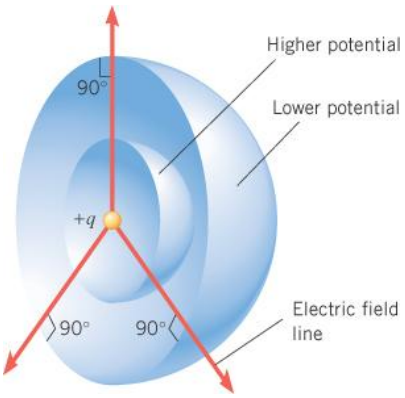
- The net electric force does <u>no work</u> for a charge moving on an equipotential surface.

$W = q\vec{\mathbf{E}} \bullet \mathbf{\Delta}\vec{\mathbf{x}} = q\left|\vec{\mathbf{E}}\right||\mathbf{\Delta}\vec{\mathbf{x}}|\cos\varphi$, where $\varphi = 90°$ Therefore W = 0

For the equipotential surfaces below, moving a charged particle from A to B to C requires no work, but moving that same charged particle from A to D does require work.



- The electric field is everywhere perpendicular to the equipotential surface.
    - The electric field must be perpendicular to the equipotential surface because if it were not there would be a component of the field along the surface which would mean there is a change in potential which violates the definition of an equipotential surface.

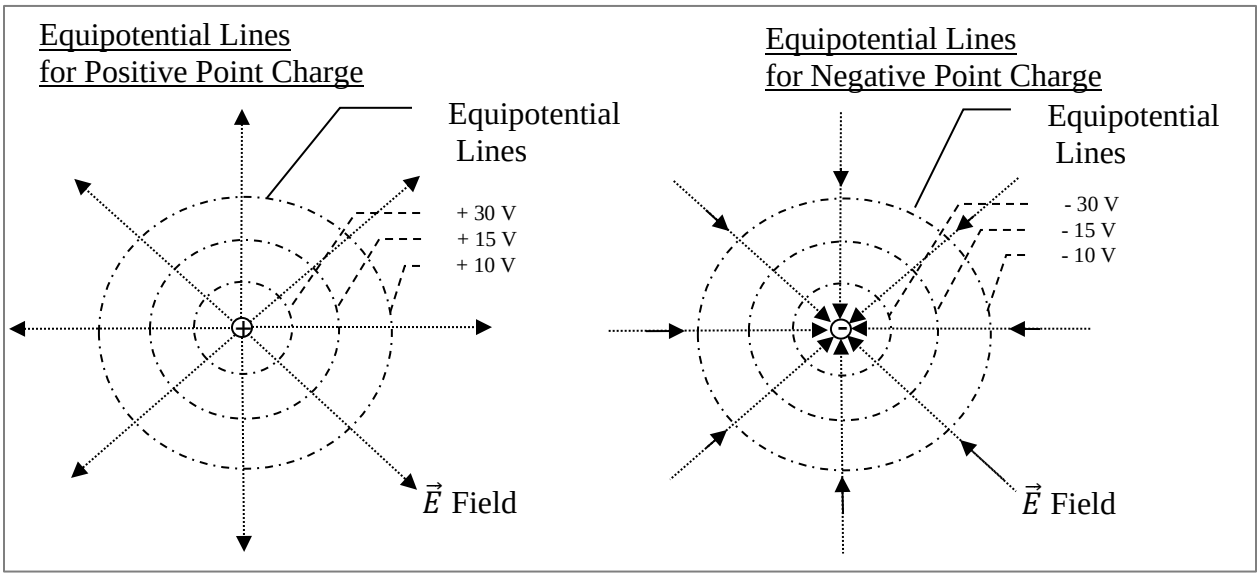


- The electric field points in the direction of decreasing potential.

$$|\vec{E}| = -\frac{\Delta V}{\Delta x}$$

- For a perfect conductor the surface is an equipotential surface because if there were a potential difference then the charge would move, and the potential would be gone.

Equipotential lines for positive and negative point charges.



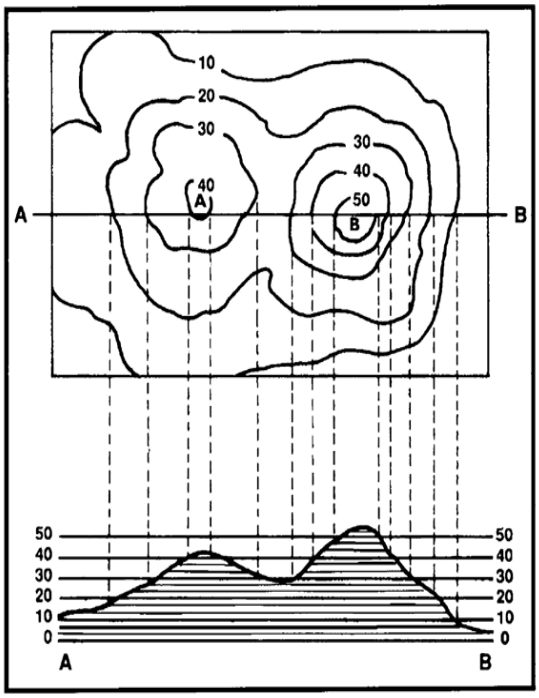
<u>Contour Map</u> – The Equipotential Lines are like contour lines or a topographical map they measure potential differences, either gravitational (mgΔy) or Electric (qΔV). By changing elevation an object changes its gravitational potential energy and by moving in an electric field (changing potentials) a charged object changes its Electric potential energy.

$$Averge\ |\vec{E}| = -\frac{\Delta V}{\Delta x} \qquad Instantanous\ |\vec{E}| = -\frac{dV}{dx}$$

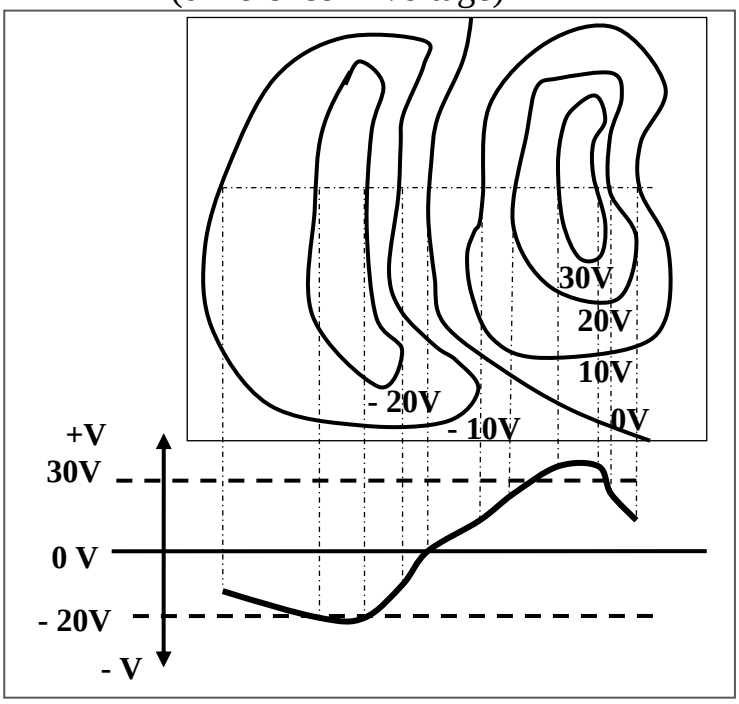
- Electric field points in the direction of decreasing potential or "downhill"
- The steepness of the slope gives the strength of the Electric Field in that region of space, the faster the potential changes the larger the field strength.

- Moving along an Equipotential line there is no change in the potential so there is no change in the charged objects' energy.
- Zero potential does not mean zero electric field.
- A positively charged object would move "downhill" from a position of high potential to a position of lower potential.

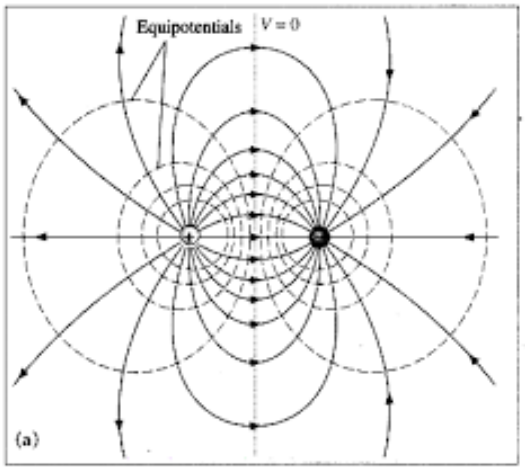
Topographical Map
(difference in height)

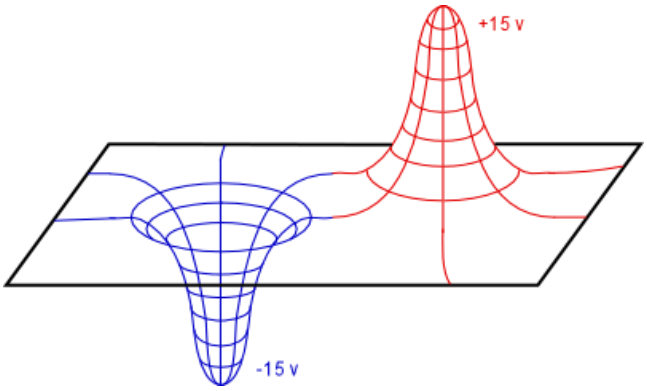


Equipotential Graph
(difference in voltage)

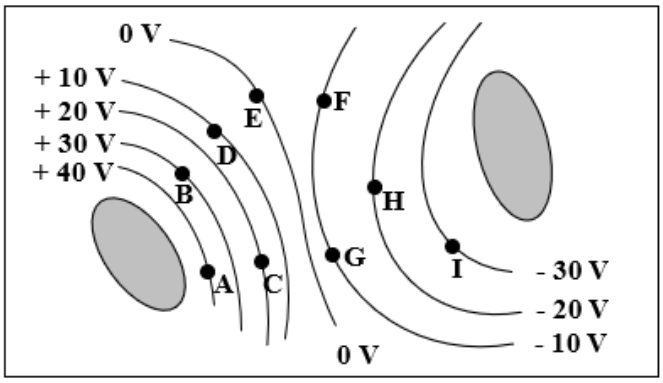


Equipotential Surfaces for a dipole - Equipotential surfaces for an electric dipole are curved, closed shapes that are always perpendicular to the electric field lines, with the central plane bisecting the dipole (zero potential) being flat and the surfaces bunching up near the charges where the field is strongest. They are symmetrical about the dipole axis and represent constant voltage, with the spacing of the equipotential lines indicating the electric field strength.





Example: The sketch shows cross sections of equipotential surfaces between two charged conductors. Points on the equipotential surfaces are labeled A, B, C, ..., I.



a) At point **D** in the diagram, what direction is the electric field pointing? The electric field goes in the direction of higher potential to lower potential, and the electric field is perpendicular to the equal potential lines therefore the electric field points in the direction of E.

b) If the spacing between the D and E is 0.2 m what is the value of the electric field between the two points?

$$\Delta V = E\Delta x \quad \text{therefore } E = \Delta V/\Delta x = 50 \text{ V/m}$$

c) Point E is at the 0 V equal potential line, is there an electric there? Yes, the electric field goes from high potential to low potential, the zero potential is just the point at which the potential goes from positive to negative.

d) How much work is required to move a positive 6.0 μC point charge from H (H is at – 20V) to G to E to F to C to B to A (+ 40 V)?

$W = \Delta PE = q\Delta V = q(V_A - V_H) = q(40 \text{ V} - (-20 \text{ V})) = (6 \times 10^{-6} \text{ C})(60 \text{ V}) = 3.6 \times 10^{-6} \text{ J}$

e) How much work is required to move a positive 6.0 μC point charge from G to F?
$W = \Delta PE = q\Delta V$ but $\Delta V = 0$ V there is no change in potential between points G and F, therefore there is no work required to move the charge W = 0 J

f) Using the diagram, If a particle of mass 0.5 g and net charge of q = + 6.0 μC is placed at point A with an initial velocity of zero and is released, how fast will it be moving when it reaches point H?

$$\Delta KE + \Delta PE_E = 0$$

Energy at point A = qV_A and Energy at Point H = $1/2mv^2 + qV_H$

Therefore the energy at A must equal the energy at H so

$qV_A = ½ mv^2 + qV_H$ and $½ mv^2 = q(V_A - V_H)$

$v = [2q(V_A - V_H)/(m)]^{1/2} = 1.2$ m/s